

The Ethical Hacker's Command Handbook

Chapter 2 – Part 6: Timing, Performance & Scan Optimization

Learn how to optimize Nmap scans for speed, reliability, and efficient network enumeration.

Learning Objectives

Understand Nmap timing templates, packet rate controls, scan delays, host timeouts, retry limits, and how to choose the right settings for different authorized environments.

Why Timing Matters

Choosing the correct timing settings balances scan speed and accuracy. Very fast scans may miss responses on slow networks, while very slow scans increase completion time.

Option	Name	Typical Use
-T0	Paranoid	Very cautious, slow scanning
-T1	Sneaky	Slow or sensitive networks
-T2	Polite	Reduce network load
-T3	Normal	Balanced default scanning
-T4	Aggressive	Most lab environments
-T5	Insane	Very fast, reliable local networks only

nmap -T0

Purpose: Paranoid timing

What it does: Slowest scan speed.

When to use: Learning cautious timing or testing slow scans.

Avoid when: Quick assessments.

Memory Tip: 0 = Slowest

nmap -T1

Purpose: Sneaky timing

What it does: Very slow scanning.

When to use: Sensitive or slow networks.

Avoid when: Routine lab work.

Memory Tip: 1 = Very slow

nmap -T2

Purpose: Polite timing

What it does: Reduces network load.

When to use: Shared or bandwidth-limited networks.

Avoid when: Time-limited scans.

Memory Tip: 2 = Polite

nmap -T3

Purpose: Normal timing

What it does: Balanced performance.

When to use: General-purpose scanning.

Avoid when: When another timing template better matches your needs.

Memory Tip: 3 = Balanced

nmap -T4

Purpose: Aggressive timing

What it does: Fast and reliable on most modern networks.

When to use: TryHackMe, Hack The Box, and local labs.

Avoid when: Unstable or high-latency networks.

Memory Tip: 4 = Fast

nmap -T5

Purpose: Insane timing

What it does: Fastest timing template.

When to use: Reliable local networks only.

Avoid when: Internet targets or unstable connections.

Memory Tip: 5 = Fastest

nmap --min-rate 1000

Purpose: Minimum packet rate

What it does: Requests at least the specified packet rate.

When to use: Controlled, fast lab environments.

Avoid when: Congested networks.

Memory Tip: Higher value = faster sending

nmap --max-rate 500

Purpose: Maximum packet rate

What it does: Limits packet sending rate.

When to use: When reducing network load.

Avoid when: When maximum speed is needed.

Memory Tip: Caps sending speed

nmap --scan-delay 500ms

Purpose: Delay between probes

What it does: Waits between packets.

When to use: Devices that respond poorly to rapid requests.

Avoid when: Speed-focused scans.

Memory Tip: Delay improves stability

nmap --host-timeout 5m

Purpose: Host timeout

What it does: Stops scanning a host after a set time.

When to use: Large assessments.

Avoid when: Short scans where timeout isn't an issue.

Memory Tip: Avoids hanging scans

nmap --max-retries 2

Purpose: Retry limit

What it does: Limits retries for unanswered probes.

When to use: Reliable lab environments.

Avoid when: Unstable networks needing higher reliability.

Memory Tip: Fewer retries = faster scans

Typical Workflow

Choose timing (-T3 or -T4) → SYN scan (-sS) → Version detection (-sV) → Default scripts (-sC) → Save results (-oA)

Revision Notes

- -T3 = Balanced
- -T4 = Best for most labs
- -T5 = Fastest
- --min-rate = Minimum packet rate
- --max-rate = Maximum packet rate
- --scan-delay = Delay between probes
- --host-timeout = Stop long scans
- --max-retries = Retry limit

Practice Questions

1. Which timing template is balanced by default?
2. When is -T4 a good choice?

3. What does `--scan-delay` control?
4. Why use `--host-timeout`?
5. How does `--max-retries` affect scanning?